

BOOK A DOCTOR SYSTEM - FRONTEND STRUCTURE

React.js frontend structure used in the Book a Doctor MERN application

```
client/
|-- public/
|   |-- favicon.svg
|   |-- icons.svg
|-- src/
|   |-- assets/
|   |   |-- hero.png
|   |   |-- react.svg
|   |   |-- vite.svg
|   |-- components/
|   |   |-- DashboardCard.jsx
|   |   |-- DoctorCard.jsx
|   |   |-- Footer.jsx
|   |   |-- Navbar.jsx
|   |   |-- Sidebar.jsx
|   |   |-- Topbar.jsx
|   |-- context/
|   |   |-- AuthContext.jsx
|   |-- pages/
|   |   |-- Home.jsx
|   |   |-- Login.jsx
|   |   |-- Register.jsx
|   |   |-- AdminDashboard.jsx
|   |   |-- DoctorDashboard.jsx
|   |   |-- PatientDashboard.jsx
|   |   |-- Doctors.jsx
|   |   |-- Patients.jsx
|   |   |-- Appointments.jsx
|   |   |-- PatientDoctors.jsx
|   |   |-- BookAppointment.jsx
|   |   |-- MyAppointments.jsx
|   |   |-- DoctorAppointments.jsx
|   |   |-- Profile.jsx
|   |-- services/
|   |   |-- api.js
|   |-- styles/
|   |   |-- BookAppointment.css
|   |   |-- Dashboard.css
|   |   |-- DashboardCard.css
|   |   |-- DoctorAppointments.css
|   |   |-- DoctorCard.css
|   |   |-- DoctorDashboard.css
|   |   |-- Doctors.css
|   |   |-- Footer.css
|   |   |-- Home.css
|   |   |-- Login.css
|   |   |-- MyAppointments.css
```

```
| | |-- Navbar.css
| | |-- PatientDashboard.css
| | |-- PatientDoctors.css
| | |-- Profile.css
| | |-- Register.css
| | |-- Sidebar.css
| | |-- Topbar.css
| |
| |-- App.jsx
| |-- App.css
| |-- main.jsx
| |-- index.css
|
|-- .gitignore
|-- eslint.config.js
|-- index.html
|-- package.json
|-- package-lock.json
|-- vite.config.js
```

Frontend Folder Responsibilities

Folder / File	Responsibility
public	Stores static files such as icons and the browser favicon.
src/assets	Contains images and visual resources used by the React interface.
src/components	Contains reusable UI components such as Navbar, Sidebar, Footer, cards, and Topbar.
src/context/AuthContext.jsx	Provides shared authentication state and user session data across React components.
src/pages	Contains full application pages for Public, Admin, Doctor, and Patient modules.
src/services/api.js	Creates the Axios instance and automatically attaches the JWT token to protected API requests.
src/styles	Contains page-specific and component-specific CSS files.
src/App.jsx	Defines all React Router routes and maps URLs to pages.
src/main.jsx	Starts the React application and renders App inside BrowserRouter.
index.html	Main HTML template used by Vite.
package.json	Stores frontend dependencies and npm scripts.
vite.config.js	Contains Vite development and build configuration.

Page Modules

Module	Pages and purpose
Public Module	Home, Login, and Register pages.
Admin Module	Admin Dashboard, Doctors, Patients, and Appointments management pages.
Patient Module	Patient Dashboard, Patient Doctors, Book Appointment, My Appointments, and Profile.

Module	Pages and purpose
Doctor Module	Doctor Dashboard, Doctor Appointments, and Profile.

Frontend Request Flow

User Action -> React Page -> Event Handler -> Axios API Request -> Express Backend -> JSON Response -> React State Update -> Updated User Interface

Technologies Used

React.js, Vite, React Router DOM, Axios, React Icons, JavaScript, HTML, and CSS.

Outcome

The frontend is organized into reusable components, role-based pages, API services, shared context, and dedicated CSS files. This structure makes the Book a Doctor interface easy to maintain, extend, and integrate with the Node.js and Express backend.